

The Human Circulatory System

Discovering How Our Heart Keeps Us Alive

Why is the Circulatory System Important? Your circulatory system is like a super delivery service working inside your body 24/7! It transports oxygen from your lungs to every cell, delivers nutrients from food to all body parts, removes carbon dioxide and waste, and keeps every organ working properly.

Part 1 — What is the Heart?

Your heart is an **amazing muscle** that works non-stop to keep you alive. It pumps blood throughout your entire body, delivering oxygen and nutrients to every part. The heart is about the **size of your fist** and sits in the middle of your chest, slightly to the left.

■ **Did you know?** Your heart beats about **100,000 times every single day** — that means it never takes a break, even when you are sleeping! It pumps about **5 litres of blood per minute**.

The Four Chambers

Right Atrium

Receives used blood from the body that needs fresh oxygen.

Right Ventricle

Pumps blood to the lungs to pick up oxygen.

Left Atrium

Receives oxygen-rich blood returning from the lungs.

Left Ventricle

The **strongest chamber!** Pumps oxygen-rich blood to your whole body.

Heart Valves

Act like **one-way doors** to keep blood moving forward and prevent backflow. They make the "lub-dub" sound!

Part 2 — Blood Vessels

Blood vessels are like **highways inside your body!** They carry blood everywhere — from your head to your toes. There are three main types:

Vessel	Function	Key Features	Largest Example
Arteries	Carry blood AWAY from heart	Thick strong walls, pulse with heartbeat	Aorta
Veins	Carry blood BACK to heart	Thinner walls, contain valves, often visible as bluish under skin	Vena Cava
Capillaries	Connect arteries to veins	Smallest (thinner than hair!); where O₂ and nutrients pass to cells	Throughout body

Part 3 — What Makes Up Blood?

Blood is made of **four main parts** that work together to keep us healthy:

Red Blood Cells

Carry oxygen from the lungs to every part of the body using haemoglobin.

White Blood Cells

Like tiny soldiers — fight germs and infections to keep us healthy.

Platelets

Help stop bleeding by clotting the blood when we get a cut.

Plasma is the watery, yellowish liquid (about 55% of blood) that carries all the blood cells around. It also transports nutrients from food to all body parts and removes waste products.

Part 4 — The Circulation Pathways

There are **two main circulation loops** that work together with every heartbeat:

Pulmonary Circulation

Blood travels from the **right side** of the heart to the **lungs** to pick up fresh oxygen. Then it returns to the left side, ready to be pumped out. This loop picks up **O₂** and releases **CO₂**.

Systemic Circulation

Oxygen-rich blood leaves the **left side** of the heart through the **aorta** and travels to every part of the body — brain, muscles, tummy, toes! Used blood returns via veins to the right side.

Part 5 — Keeping Your Heart Healthy

■ Eat Healthy Foods	Fill your plate with colourful fruits, vegetables, whole grains, and lean proteins.
■ Exercise Regularly	Run, play, dance or swim for at least 60 minutes every day to keep your heart muscle strong.
■ Drink Plenty of Water	Water helps your blood flow smoothly and keeps your whole body working well.
■ Get Enough Sleep	Your heart rests and repairs while you sleep — aim for 9-11 hours every night.
■ Avoid Too Much Junk Food	Too much salt, sugar, and fat can make your heart work harder over time.

Part 6 — Fun Facts!

♥■ **100,000 beats/day**
Your heart never stops — it works even when you sleep!

■ **5 L/min pumped**
That's like filling 5 big water bottles every single minute!

■ **100,000 km of vessels**
Your blood vessels end-to-end would wrap around the Earth twice!

Part 7 — Practice Questions

	Question	Answer
Q1	What is the main job of the heart?	<i>To pump blood throughout the entire body</i>
Q2	Name the 4 chambers of the heart.	<i>Right atrium, right ventricle, left atrium, left ventricle</i>
Q3	What do arteries do?	<i>Carry blood away from the heart</i>
Q4	What do veins do?	<i>Carry blood back to the heart</i>
Q5	Name the 4 components of blood.	<i>Red blood cells, white blood cells, platelets, plasma</i>

Q6	What is pulmonary circulation?	<i>The loop from the heart to the lungs to pick up oxygen</i>
Q7	What is the largest artery?	<i>The aorta</i>
Q8	Name 3 ways to keep your heart healthy.	<i>Exercise, eat well, sleep enough, drink water, avoid junk food</i>

Key Takeaways

	Heart	A fist-sized muscle pumping blood 100,000 times daily through 4 chambers
	Arteries	Carry oxygen-rich blood AWAY from the heart; thick walls; largest = aorta
	Veins	Carry used blood BACK to the heart; thinner walls; contain valves
	Capillaries	Tiny connectors where oxygen and nutrients pass into cells
	Blood	4 parts: red cells (O ₂), white cells (immunity), platelets (clotting), plasma (transport)
	Pulmonary	Right heart → lungs → left heart: picks up fresh oxygen
	Systemic	Left heart → whole body → right heart: delivers O ₂ and nutrients

Keep exploring the wonders of science!
Your heart is beating right now — amazing, isn't it?